DARPA-PS-26-09: Strategic Technology Office (STO) Office-Wide ISO

PROPOSAL ABSTRACT — TITLE PAGE

Inversion-Free, Machine-Verified Algebraic Null Steering for CRPA Anti-Jamming: Provable Leakage Bounds, Phase-Continuous Null Switching, and Fixed-Point RFSoC Validation

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Solicitation / PM:	DARPA-PS-26-09 (STO)
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CAGE / NCAGE Code:	Foreign entity — NCAGE registration pending
Proposed ROM & Period:	$450,000 USD (12 months)

SETA / A&AS Non-Conflict Statement: No member of the proposing team works for DARPA as SETA, A&AS, or similar support-services personnel.

Applicability Statements:

1. **STO technical areas:** Resilient PNT in contested electromagnetic environments; low-SWaP phased-array signal processing for attritable platforms.
2. **Portfolio relevance:** A CRPA null-steering stage with no matrix inversion, no DSP48 multipliers, single-cycle latency, and formally verified behavior, suited to platforms that cannot host a floating-point STAP processor.
3. **Advance over state of practice:** Replaces iterative/inverse-based weight computation with a closed-form algebraic construction whose two key guarantees—bounded leakage under signature error and zero carrier-phase discontinuity across null switches—are theorems checked in Lean 4, not simulation claims.
4. **Work not yet completed:** The algebra, the leakage bound, and the simulation regime map exist. Fixed-point RTL, calibration under mutual coupling, and live-RF validation do not.
5. **No prior government funding** for this concept.

1. Problem and What We Claim

Controlled Reception Pattern Antennas (CRPAs) on tactical GNSS receivers null jammers with MVDR/STAP weights $\mathbf{w} \propto \hat{\mathbf{R}}_{xx}^{-1}\mathbf{a}_{\text{sat}}$, which require a covariance estimate over K snapshots and an $\mathcal{O}(N^3)$ inversion. This is the right tool when snapshots are plentiful: with $K \geq$ (number of interferers), our simulations show MVDR reaching 85–95 dB null depth against a $+60$ dB JNR jammer and *outperforming every signature-driven nuller*, including ours. We do not claim otherwise.

What MVDR does not provide is (i) a weight update that is closed-form, inversion-free and representable in 16-bit fixed point with no multipliers; (ii) any guarantee about carrier-phase continuity when nulls are re-steered; (iii) adequate null depth when K falls below the interferer count (at $K = 2$ against 3 and 6 jammers we measure 21 dB and 10 dB respectively, versus 68 dB and 65 dB for a signature-driven nuller). The proposed effort delivers a null-steering stage that provides (i)–(iii), with its limits stated and proven rather than hidden.

2. Technical Approach

2.1 Algebraic construction

An 8-element complex baseband snapshot ($\mathbb{C}^8 \cong \mathbb{R}^{16}$) is treated as an element of the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S}$. Unlike $\mathbb{C}, \mathbb{H}, \mathbb{O}$, $\mathbb{S}$ has zero divisors: for a jammer signature u mapped onto the Moreno cone, the left-multiplication operator $L(u)$ has an exact 4-dimensional kernel with a *closed-form* basis $v = (c,\ (ac)b)$ (Theorem 5.1 of [1], machine-checked in Lean 4/Mathlib). The nulling weight is obtained by a handful of octonion products—no covariance, no inversion, no iteration. Two structural properties follow and are both machine-checked:

- **Anti-partner gain:** any satellite signature in the orthogonal anti-partner direction is passed with exactly $+3.01$ dB coherent gain (measured $\sigma = 0$ over 256 trials).
- **Phase-continuous null switching:** the 15 Fano hyperplanes of $\mathrm{PG}(3,2)$ pairwise intersect in a quaternion subalgebra $\mathbb{H}_{L_{12}}$ on which the associator vanishes identically (`gnss_pll_hinge_associative`, checked over all 56 workspace transitions). Routing the carrier reference through the hinge gives 0.0000° phase jump at a null switch; the unrouted sedenion filter gives 58° and MVDR re-estimation 5.8° with a 10% cycle-slip rate.

2.2 The leakage bound: what null depth is actually achievable

A nuller built from an estimated signature $\hat{u}$ sees the true signature $u = \hat{u} + \delta$. We prove (Lean 4, theorem `leakage_bound`, standard Mathlib axioms only):

$$\hat{u} \cdot v = 0 \implies \|(\hat{u} + \delta) \cdot v\|_2^2 \ \leq\ 2\,\|\delta\|_2^2\,\|v\|_2^2.$$

Null depth is therefore governed by signature-estimation error, $\approx 20\log_{10}(1/\|\delta\|) - 3$ dB, and *no algebraic structure can beat this*. The measured sedenion residual sits within $1.4\times$ of the bound at every tested $\|\delta\|$ (Table 1); per output beam it matches an orthogonal-projection nuller built from the same $\hat{u}$ to within 2 dB. Figures such as "139 dB" or "312 dB" reported in our earlier simulations are the $\delta = 0$ floating-point floor and are not performance claims.

2.3 Operating regime

Per-channel gain/phase mismatch of $(1\%, 1^\circ)$ yields $\|\delta\| \approx 0.02$ and a $\sim$34 dB (16-vector) / $\sim$50 dB (per-beam) null; $(0.1\%, 0.1^\circ)$ yields $\sim$54 / $\sim$70 dB. Hence the method is calibration-

Table 1: Measured null depth vs. signature error $\|\delta\|$ (8-element $\lambda/2$ ULA, 256 trials each; `bench/gps_jamming/run_robustness.py`).

$\|\delta\|$	Lean bound	Sedenion (16-vector)	Sedenion (per beam)	Orth. projection	Bound violations
10^{-3}	57.0 dB	60.1 dB	75.6 dB	77.8 dB	0/256
10^{-2}	37.0 dB	40.1 dB	55.2 dB	57.1 dB	0/256
10^{-1}	17.0 dB	20.1 dB	—	37.4 dB	0/256

limited, like all signature-driven nullers, and the proposed work is correspondingly centered on calibration and fixed-point realization rather than on depth claims. The regime where it is the right choice: fast interferer dynamics or dense multi-jammer scenes where $K < J$, SWaP budgets that exclude a floating-point inversion, and RTK/carrier-phase receivers where null re-steering must not slip the PLL.

3. 12-Month Plan and Deliverables

1. **Task 1 (Mo. 1–4) — Calibration to the leakage bound.** Closed-loop pilot-tone and blind calibration of the $\mathbb{C}^8 \to \mathbb{S}$ map under measured mutual coupling; target $\|\delta\| \le 3 \times 10^{-3}$ ($\ge$50 dB per-beam null) across L1/L2. *D1:* calibration algorithm and error budget tied to the proven bound.
2. **Task 2 (Mo. 3–8) — Fixed-point RTL.** 16-bit Q14 pipeline on Zynq UltraScale+ RFSoC implementing the cone projection, nullspace partner, and 35 hinge projections; our Q14 model shows 0/256 sign-table mismatches and $\sim$75 dB quantization-limited depth with no DSP48 usage. *D2:* SystemVerilog IP, bit-true test suite, and a Lean-to-RTL conformance check of the sign table.
3. **Task 3 (Mo. 7–12) — Live-RF HIL.** 8-channel phase-coherent SDR testbed with a GNSS constellation simulator under wideband, chirp and repeater interference; MVDR/LSMI run side-by-side on the same captures. *D3:* data package, final report, and a published regime map (null depth vs. K, J, $\|\delta\|$) for both methods.

Risk: the principal risk is that achievable $\|\delta\|$ on real hardware is $> 10^{-2}$, capping per-beam depth near 50 dB. This is a known, bounded outcome, not a failure mode; the Task 3 regime map is the deliverable either way. **Transition:** GPR on all RTL and calibration code; integration path via a U.S. prime for M-code CRPA retrofits. The proposer is a foreign entity and will team with a U.S. organization as required for any follow-on involving controlled technical data.

References: [1] D. C. Agourakis, "Sedenion Algebras and PG(3,2) Incidence Structures…," Zenodo 10.5281/zenodo.23588274 (Lean 4 sources: `github.com/josaum/sedenion-pg32`). [2] D. C. Agourakis, "Sedenion PG(3,2) Zero-Divisor Beamforming for GNSS Anti-Jamming," Zenodo 10.5281/zenodo.23603006.

Price Realism / Rough Order of Magnitude (ROM)

Total ROM for the 12-month effort: **$450,000 USD** (range $420k–$475k depending on RF testbed rental). Prior algebraic research and Lean 4 verification are leveraged at no cost to the Government.

Category	**Basis (12 mo.)**	**USD**
Principal Investigator / algorithm & calibration lead	1,200 h @ $150/h loaded	180,000
Senior FPGA/RFSoC RTL engineer	960 h @ $135/h loaded	129,600
RF / GNSS test engineer (HIL execution)	500 h @ $110/h loaded	55,000
RFSoC dev kit, phase-matched cabling, 8-element fixture, GNSS/EW generator rental	quotes	65,400
Travel and DoD reviews (2 trips, Arlington VA)		20,000
Total	**2,660 h + materials**	**450,000**